

Here is Newton's decisive stroke: draw QR PARALLEL to the radius SP rather than straight to the tangent, and this parallel is one heart of the construction. Then drop QT PERPENDICULAR to SP , the second heart, giving a right angle at T . Because $QR \parallel SP$, the versed sine now speaks directly in the language of the force's own direction. And because $QT \perp SP$, the triangle SQP has area $\frac{1}{2} SP \cdot QT$ — which, by Prop. I, is proportional to the time in which the arc is swept.